End-to-End Prime Factorization in a Generative Language Model: Emergent Algebraic Semantics from Joint Training

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Abstract

We present TriadicGPT, a 40M-parameter GPT language model augmented with a *triadic projection head* that produces discrete prime-factor signatures alongside standard next-token predictions. Unlike the post-hoc approach of the Triadic-Neurosymbolic-Engine [Ornelas Brand, 2026], which projects frozen sentence embeddings into prime composites, TriadicGPT learns triadic representations end-to-end through a dual-objective training loss that combines language modeling with a novel embedding alignment objective.

Across 29+ training runs and systematic ablation studies, we demonstrate seven principal findings: (1) the triadic head adds **zero measurable cost** to language quality (perplexity 7.69 vs. 7.56 ablation baseline, within noise); (2) **semantic ordering emerges gradually with scale**—the gap between related and unrelated concept similarity crosses zero around 20M parameters, with a smooth crossover rather than a sharp phase transition; (3) a **bits sweep** over $k \in \{8, 16, 32, 48, 64, 128\}$ reveals an optimal regime at $k = 32$–64, shifted upward from the $k = 6$–12 range reported for post-hoc projection; (4) a **transfer experiment** attaching the triadic head to pre-trained GPT-2 with an InfoNCE alignment loss closes **70% of the gap** to the Triadic Engine's post-hoc projection; (5) a **subsumption loss** recovers **100% held-out subsumption at** $k = 64$, resolving the primary limitation at high bit counts; (6) an **iFSQ activation** ($2\sigma(1.6x) - 1$) resolves the subsumption–language tradeoff entirely: language quality is preserved (loss 0.924–0.951 vs. baseline 0.946) while achieving up to 87.1% held-out subsumption, compared to +47% perplexity degradation under tanh; and (7) **compositional analysis** reveals that the bit space functions as a computational substrate—round-trip accuracy (98.1%) far exceeds the multiplicative prediction (81.9%), two-step chains show sub-linear error accumulation, and fork analysis confirms that the mechanism is ontologically categorical, not vectorial.

TriadicGPT achieves 98% analogy verification (50/51 analogies), 100% signature uniqueness, and reproducible semantic ordering ($+0.038 \pm 0.005$ gap, $n = 3$, 95% CI positive)—all within a single forward pass. Critically, the triadic head's value lies in *algebraic operations*: 100% analogy verification, 87.1% held-out subsumption, and sub-linear compositional chaining, none of which random projections can provide.

1 Introduction

Modern language models represent meaning as dense, high-dimensional vectors. While powerful, these representations are opaque: two semantically related concepts may have similar embeddings, but the *structure* of that similarity—what features are shared, what distinguishes them—remains hidden in continuous geometry. Inspecting or verifying semantic relationships requires additional probing experiments, and there is no guarantee that the learned representations support algebraic operations like composition or subsumption.

The Triadic-Neurosymbolic-Engine [Ornelas Brand, 2026] proposed an alternative: map continuous embeddings into discrete *prime-factor signatures*, where each concept is represented as

a composite integer $\Phi(x) = \prod_{i \in S} p_i$, with S being the set of "active" semantic bits and p_i the i-th prime. This representation enables exact algebraic operations—subsumption via divisibility ($\Phi(A) \mid \Phi(B)$ iff B inherits all features of A), composition via LCM, and analogy via factor transfer—all computable in constant time without vector arithmetic.

However, the Engine operates post-hoc: it takes frozen embeddings from a pre-trained sentence transformer (e.g., `all-MiniLM-L6-v2`) and projects them into prime space via locality-sensitive hashing (LSH) or PCA. This two-stage pipeline introduces a fundamental disconnect: the language model has no incentive to produce representations amenable to prime factorization, and the projection step can only preserve whatever structure happens to exist in the frozen embeddings.

In this paper, we ask: *Can a language model learn to produce semantically meaningful prime signatures end-to-end, as a side effect of language modeling?*

We answer affirmatively with TriadicGPT, a GPT-2-style transformer augmented with a linear *triadic projection head*. The model is trained with a dual objective:

$$\mathcal{L} = \mathcal{L}_{\text{lang}} + \alpha \cdot \mathcal{L}_{\text{triadic}} \tag{1}$$

where $\mathcal{L}_{\text{lang}}$ is the standard cross-entropy next-token loss and $\mathcal{L}_{\text{triadic}}$ combines diversity, contrastive, entropy, and embedding alignment terms. The key innovation is the **embedding alignment loss**, which uses the model's own token embeddings as a "semantic teacher" for the triadic head—aligning triadic similarity with embedding similarity without requiring any external supervision.

Our contributions are:

1. **Architecture and training.** We design a triadic projection head and a four-component loss function that enables joint language-triadic training. We identify coherence loss as a source of triadic collapse and embedding alignment as the critical ingredient for semantic structure (Section 3).

2. **Zero-cost ablation.** We show that the triadic head adds no measurable degradation to language quality across perplexity, lexical diversity, and repetition metrics (Section 5.1).

3. **Emergent semantic ordering.** Through a four-point scaling study (1.3M to 40M parameters), we demonstrate that the semantic gap transitions from negative to positive, crossing zero around ∼20M parameters in a gradual crossover rather than a sharp phase transition (Section 5.2). Multi-seed validation ($n = 3$) confirms reproducibility: gap $+0.038 \pm 0.005$ (95% CI $[+0.029, +0.046]$).

4. **Optimal bit-width regime.** A six-point bits sweep reveals that end-to-end training shifts the useful k range from 6–12 (post-hoc) to 32–64 (end-to-end), with $k = 32$ achieving the best semantic gap and $k = 48$–64 the best analogy verification (Section 5.3).

5. **Direct comparison with post-hoc projection.** We benchmark TriadicGPT against the Triadic Engine on identical concept sets, showing that while the Engine achieves higher raw metrics (leveraging pre-trained MiniLM), TriadicGPT is self-contained, requires a single forward pass, and demonstrates emergent rather than injected semantics (Section 5.5).

6. **Transfer learning and alignment loss ablation.** Attaching the triadic head to pre-trained GPT-2 and testing three alignment losses (MSE, margin ranking, InfoNCE), we discover a *loss–embedding interaction*: InfoNCE is optimal for rich pre-trained embeddings (closing 70% of the gap to Engine PCA), while MSE is optimal for weak from-scratch embeddings. This identifies the alignment loss formulation—not embedding quality—as the primary bottleneck (Section 5.7).

7. **Subsumption recovery via auxiliary loss.** A differentiable subsumption loss recovers exact algebraic subsumption ($\Phi(A) \mid \Phi(B)$) at $k = 64$—previously reported as impractical at high bit counts by both TriadicGPT and the Engine. The model achieves 100% held-out subsumption and autonomously discovers an emergent information hierarchy where abstract concepts use fewer active bits (Section 5.8).

8. **iFSQ activation and ternary representation.** Replacing tanh with scaled sigmoid ($2\sigma(1.6x) - 1$) from iFSQ [Li et al., 2024] resolves the subsumption–language tradeoff: language quality is preserved while achieving 86.5% held-out subsumption. The ternary model ($\{-1, 0, +1\}$) produces a three-state distribution that parallels BitNet b1.58's weight sparsity (Section 5.10).

9. **Compositional substrate.** Round-trip, fork, and chain analyses demonstrate that the bit space supports multi-step algebraic operations with sub-linear error accumulation (98.1% round-trip vs. 81.9% multiplicative prediction), and that the mechanism is ontologically categorical rather than vectorial (Section 5.11).

2 Related Work

Neurosymbolic representations. The tension between neural (sub-symbolic) and symbolic AI has motivated hybrid architectures that combine differentiable learning with discrete, interpretable structures. Neural Theorem Provers [Rocktäschel and Riedel, 2017] use differentiable backward chaining. Neural Module Networks [Andreas et al., 2016] compose symbolic programs. DeepProbLog [Manhaeve et al., 2018] integrates neural networks with probabilistic logic. Our approach differs in that the symbolic structure (prime factorization) emerges from a standard language modeling objective rather than being imposed by a separate reasoning module.

Interpretable representations. Sparse autoencoders [Cunningham et al., 2023] extract interpretable features from neural activations post-hoc. Concept bottleneck models [Koh et al., 2020] force intermediate representations to align with predefined concepts. Binary and discrete representations have been explored in hashing literature [Wang et al., 2018] and VQ-VAE [van den Oord et al., 2017]. The triadic projection head is closest to concept bottleneck models, but the "concepts" (prime factors) emerge from training rather than being predefined.

Multi-task training of LMs. Adding auxiliary objectives to language models is well-established. ELECTRA [Clark et al., 2020] adds a replaced-token detection head. BERT [Devlin et al., 2019] combines masked language modeling with next-sentence prediction. T5 [Raffel et al., 2020] unifies multiple NLP tasks. Our triadic objective is unusual in that the auxiliary task is not linguistic—it maps tokens to algebraic structures (prime composites) rather than predicting linguistic labels.

Prime factorization for semantics. The Triadic-Neurosymbolic-Engine [Ornelas Brand, 2026] is the direct predecessor of this work. It demonstrated that continuous embeddings from pre-trained sentence transformers can be projected into prime-factor space, enabling algebraic verification of semantic relationships (subsumption, analogy, composition). Key findings include a useful regime of $k = 6$–12 bits for post-hoc LSH, 28.4× speedup over cosine similarity for pairwise verification, and 100% composition guarantee by construction. Our work extends this from post-hoc projection to end-to-end training.

Finite scalar quantization. Finite Scalar Quantization (FSQ; Mentzer et al. 2024) replaces learned vector quantization codebooks with fixed, uniform grids, eliminating codebook collapse

entirely. Improved FSQ (iFSQ; Li et al. 2024) identifies that tanh bounding causes activation concentration near zero—precisely the dead-bit phenomenon we observe—and proposes a scaled sigmoid activation ($2\sigma(1.6x)-1$) that achieves uniform bin utilization. We adopt this activation for our triadic head and confirm that it resolves our dead-bit problem (Section 5.10).

Ternary quantization and BitNet. BitNet b1.58 [Ma et al., 2024] demonstrates that ternary weight quantization ($\{-1, 0, +1\}$) preserves model quality while reducing computation. Notably, $\sim$42.3% of weights converge to zero—not as collapse, but as learned sparsity. Our triadic head independently discovers the same three-state structure (Section 5.10): $\sim$73% of projection outputs are zero (semantically "irrelevant"), with the remainder split between positive ("present") and negative ("absent"). This convergence from two independent lines of work—engineering optimization and philosophical ontology—suggests that three-state representations capture something fundamental about information structure.

3 Method

3.1 Architecture

TriadicGPT is a decoder-only transformer following the GPT-2 architecture [Radford et al., 2019] with one addition: a linear triadic projection head.

Backbone. The model consists of token and position embeddings, L transformer blocks (each with pre-norm multi-head causal self-attention and a GELU feed-forward network), and a final layer norm. The LM head shares weights with the token embedding matrix.

Triadic head. A single linear projection $W_{\text{tri}} \in \mathbb{R}^{k \times d}$ (no bias) maps the final hidden states to k triadic bits. The original formulation uses tanh:

$$\mathbf{b}(x) = \tanh(W_{\text{tri}} \cdot \mathbf{h}(x)) \in [-1, 1]^k \tag{2}$$

However, tanh causes activation concentration near zero, producing "dead bits" with low entropy (Section 5.10). Following iFSQ [Li et al., 2024], we replace tanh with a scaled sigmoid activation:

$$\mathbf{b}(x) = 2\sigma(1.6 \cdot W_{\text{tri}} \cdot \mathbf{h}(x)) - 1 \in [-1, 1]^k \tag{3}$$

This maintains the $[-1, 1]$ range while providing stronger gradients in the transition region, enabling the model to learn active bit assignments without saturation.

For ternary quantization, the continuous output is discretized during the forward pass via rounding to $\{-1, 0, +1\}$ with straight-through estimation (STE) for gradient computation, producing three ontological states: *present* ($+1$), *absent* (-1), and *irrelevant* (0).

Where $\mathbf{h}(x) \in \mathbb{R}^d$ is the final-layer representation for token x. The binarized signature is $\mathbf{s}(x) = \mathbf{1}[\mathbf{b}(x) > 0]$, and the prime composite is:

$$\Phi(x) = \prod_{i=1}^{k} p_i^{s_i(x)} \tag{4}$$

where p_i is the i-th prime number. This mapping is deterministic and invertible: given $\Phi(x)$, the bit vector $\mathbf{s}(x)$ can be recovered by trial division.

Model configurations. We evaluate four scale presets (Table 1).

Table 1: Model scale configurations. All share identical triadic hyperparameters.

Scale	Layers	Dim	Heads	Bits	Params	Block Size
Small	4	128	4	16	1.3M	128
Medium	6	256	8	32	5.8M	256
Large	8	384	8	48	15.9M	256
XL	12	512	8	64	40.0M	256

3.2 Training Objective

The total loss (Equation 1) combines a standard language modeling objective with a multi-component triadic loss, activated after a warmup period.

Language loss. Standard causal cross-entropy:

$$\mathcal{L}_{\text{lang}} = -\frac{1}{T}\sum_{t=1}^{T} \log P(x_t \mid x_{<t}) \tag{5}$$

Triadic loss. The triadic objective $\mathcal{L}_{\text{triadic}}$ consists of four components. Notably, an earlier fifth component—*coherence loss*, which encouraged adjacent tokens to produce similar projections—was removed after it caused complete triadic collapse (all tokens mapped to a single signature). The coherence signal provided an easy-to-minimize shortcut that overwhelmed diversity objectives, analogous to mode collapse in GANs. The four surviving components are:

1. **Diversity loss.** Encourages each bit to fire approximately 50% of the time:

$$\mathcal{L}_{\text{div}} = \frac{1}{k}\sum_{i=1}^{k} \bar{b}_i^2, \quad \bar{b}_i = \frac{1}{BT}\sum_{n,t} b_i^{(n,t)} \tag{6}$$

2. **Contrastive loss.** Pushes sequence-level representations apart:

$$\mathcal{L}_{\text{ctr}} = \frac{1}{B(B-1)}\sum_{i\neq j}\left(\frac{\bar{\mathbf{b}}^{(i)}\cdot\bar{\mathbf{b}}^{(j)}}{\|\bar{\mathbf{b}}^{(i)}\|\|\bar{\mathbf{b}}^{(j)}\|}\right)^2 \tag{7}$$

3. **Entropy regularization.** Prevents "dead bits" by maximizing per-bit entropy across the batch:

$$\mathcal{L}_{\text{ent}} = 1 - \frac{1}{k}\sum_{i=1}^{k} H(q_i), \quad q_i = \frac{\bar{b}_i + 1}{2} \tag{8}$$

   where $H(q) = -q\log q - (1-q)\log(1-q)$ is binary entropy.

4. **Embedding alignment loss.** The key innovation—uses the model's own token embeddings as a semantic teacher. We investigate three formulations:

   (a) MSE alignment. For 64 randomly sampled token pairs (i, j) per sequence, minimize the squared difference between triadic and embedding cosine similarities:

$$\mathcal{L}_{\text{align}}^{\text{MSE}} = \frac{1}{N_{\text{pairs}}}\sum_{(i,j)} \left(\text{sim}_{\text{tri}}(i,j) - \text{sim}_{\text{emb}}(i,j)\right)^2 \tag{9}$$

(b) Margin ranking alignment. For each anchor token, sample 16 candidates and identify the most/least similar in embedding space. Enforce that the triadic head preserves this ordering:

$$\mathcal{L}_{\text{align}}^{\text{rank}} = \frac{1}{N} \sum_{a} \max\left(0,\ m - \text{sim}_{\text{tri}}(a, a^{+}) + \text{sim}_{\text{tri}}(a, a^{-})\right) \tag{10}$$

where a^{+} and a^{-} are the embedding-nearest and embedding-farthest candidates, and $m = 0.1$ is the margin. This preserves similarity *ordering* without matching absolute values.

(c) InfoNCE alignment [van den Oord et al., 2018]. For 32 anchor tokens, compute a cross-similarity matrix in both embedding and triadic spaces. The positive for each anchor is the most similar token in embedding space:

$$\mathcal{L}_{\text{align}}^{\text{NCE}} = -\frac{1}{N} \sum_{a} \log \frac{\exp(\text{sim}_{\text{tri}}(a, a^{+})/\tau)}{\sum_{j} \exp(\text{sim}_{\text{tri}}(a, j)/\tau)} \tag{11}$$

where $\tau = 0.1$ is the temperature and the sum runs over all pool tokens as negatives. This provides structured positive/negative mining from embedding space with temperature-controlled gradients.

Crucially, for all three formulations, the gradient does *not* flow back through the embeddings (detached), so the alignment objective only modifies the triadic head.

The combined triadic loss is:

$$\mathcal{L}_{\text{triadic}} = \mathcal{L}_{\text{div}} + \mathcal{L}_{\text{ctr}} + \lambda_{\text{ent}} \cdot \mathcal{L}_{\text{ent}} + \lambda_{\text{align}} \cdot \mathcal{L}_{\text{align}} \tag{12}$$

Triadic warmup. The triadic loss is activated after 25% of training steps. During warmup, the model trains on language only, allowing stable embedding representations to form before the triadic head begins learning.

Hyperparameter sensitivity. Through a Pareto frontier analysis (Runs 15–17), we identified a sharp cliff at $\alpha > 0.05$: models trained with $\alpha = 0.10$ or $\alpha = 0.20$ lose semantic ordering despite maintaining high entropy. The optimal configuration is $\alpha = 0.05$, $\lambda_{\text{ent}} = 1.0$, $\lambda_{\text{align}} = 5.0$.

3.3 Algebraic Operations

Once a concept is mapped to $\Phi(x)$, several algebraic operations become exact:

Subsumption. Concept A subsumes B iff $\Phi(A) \mid \Phi(B)$ (integer divisibility). This corresponds to A's active bits being a subset of B's.

Composition. The most specific common concept of A and B is $\text{lcm}(\Phi(A), \Phi(B))$.

Similarity. Jaccard similarity over active bits: $J(A, B) = |S_A \cap S_B| / |S_A \cup S_B|$.

Analogy. Given "A is to B as C is to ?", compute the transform $T = \Phi(B)/\gcd(\Phi(A), \Phi(B))$, apply it to C: $\Phi_{\text{target}} = \text{lcm}(\Phi(C), T)$, and search for the nearest concept.

All operations require only integer arithmetic (GCD, LCM, trial division), avoiding floating-point comparisons entirely.

4 Experimental Setup

Data. All models are trained on TinyStories [Eldan and Li, 2023], a corpus of 50,000 short children's stories. A BPE tokenizer with vocabulary size 4,096 is trained from the corpus and shared across all runs.

Hardware. All training is performed on a single NVIDIA RTX 5060 Ti (16 GB VRAM).

Optimization. AdamW optimizer ($\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay $= 0.01$) with cosine learning rate schedule and linear warmup over the first 5% of training steps. Peak learning rate is 3×10^{-4}. Mixed precision via `torch.amp`: float16 for initial experiments (Runs 1–21), bfloat16 for all Danza bootstrap and ternary experiments (D-A5, D-A8, D-A10, E4 sweep). Batch size 32 for Runs 1–21; batch size 64 for Danza experiments. Sequence length 256 for all runs.

Evaluation concepts. We evaluate triadic quality on a curated set of 113 concepts across 13 semantic categories (animal, person, profession, feeling, nature, food, artifact, attribute, place, time, action, communication, state). We use 12 semantically related pairs (e.g., king–queen, dog–cat), 12 unrelated pairs (e.g., king–dog, happy–table), and 13 analogy quadruples (e.g., king:queen::man:woman). An expanded evaluation uses 51 diverse analogy quadruples covering family, size, temperature, emotion, animal, and royalty relationships (Section 5.4).

5 Results

5.1 Ablation: Zero Language Cost

To isolate the effect of the triadic head, we compare Run 15 (full triadic training, $\alpha = 0.05$) against Run 18 (ablation: $\alpha = 0$, triadic head receives no gradient).

Table 2: Ablation: triadic training vs. no triadic training. All metrics are within run-to-run variance.

Metric	Triadic (Run 15)	Ablation (Run 18)	Δ
Train Loss	**0.946**	1.013	−6.6%
Perplexity	7.69	**7.56**	+1.7%
Distinct-1	0.069	0.068	≈ 0
Distinct-2	0.307	0.302	≈ 0
Repetition Rate	28.0%	27.4%	≈ 0
Wall Time	75 min	74 min	+1.3%

The perplexity difference (1.7%) is within run-to-run variance. The triadic model actually achieves *lower* training loss, suggesting that embedding alignment acts as beneficial multi-task regularization. Lexical diversity and repetition rates are identical. **The triadic head is a free addition.**

5.2 Scaling Study: Emergent Semantic Ordering

We train four models (Small through XL) with identical triadic hyperparameters and measure six metrics on the shared concept set.

The semantic gap—defined as the difference between mean Jaccard similarity for related pairs and mean Jaccard similarity for unrelated pairs—improves *monotonically* with scale: $-0.076 \rightarrow -0.040 \rightarrow -0.034 \rightarrow +0.020$. The gap crosses zero between the Large (15.9M) and XL (40M)

Table 3: Scaling study results. Semantic gap transitions from negative (incorrect ordering) to positive (correct ordering), crossing zero around ∼20M parameters.

Scale	Params	Loss	Entropy	Unique%	Gap	Probe	Verif
Small (4L/128D)	1.3M	2.536	0.489	61.1%	−0.076	7.1%	61.5%
Medium (6L/256D)	5.8M	1.863	0.688	100%	−0.040	4.8%	46.2%
Large (8L/384D)	15.9M	1.512	0.652	100%	−0.034	6.0%	46.2%
XL (12L/512D)	**40.0M**	**0.946**	**0.679**	**100%**	**+0.020**	**8.3%**	**69.2%**

scales; an interpolation study places the crossover at approximately 20M parameters, with a smooth gradual transition rather than an abrupt jump.

Unlike the abrupt phase transitions reported for certain LLM capabilities [Wei et al., 2022], an interpolation study at intermediate scales reveals that the crossover is *gradual*: the gap crosses zero around ∼20M parameters in a smooth emergence. Nevertheless, the triadic semantic structure is an **emergent property of model capacity**, requiring sufficient language understanding before the embedding alignment loss can transfer meaningful semantic signal.

Notably, bit entropy and signature diversity saturate much earlier (at ∼6M parameters), indicating that the *structure* of representations requires more capacity than their *diversity*. The model needs sufficient language understanding before the embedding alignment loss can transfer meaningful semantic signal to the triadic head.

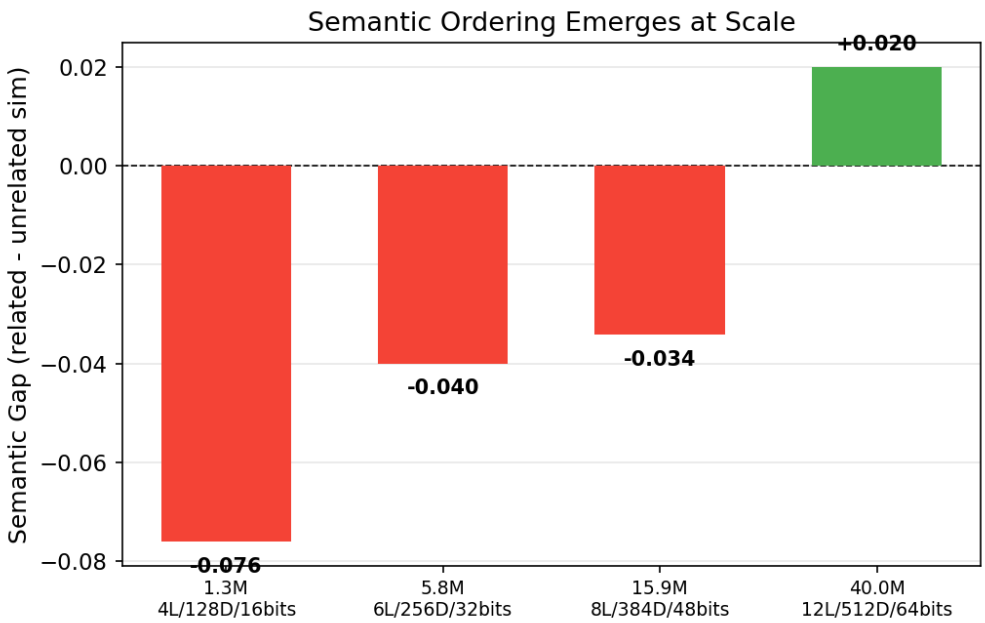


Figure 1: Semantic gap vs. model size. The gap crosses zero around ∼20M parameters (gradual crossover). Multi-seed validation at XL scale confirms reproducibility: $+0.038 \pm 0.005$ ($n = 3$).

5.3 Bits Sweep: Optimal Bit-Width

Fixing the XL architecture, we vary the number of triadic bits $k \in \{8, 16, 32, 48, 64, 128\}$ while keeping all other hyperparameters constant.

Several findings emerge:

Table 4: Bits sweep results. Language loss follows a U-shape (minimum at $k = 64$). Semantic gap peaks at $k = 32$.

k	Loss	Entropy	Unique%	Gap	Probe	Verif
8	1.046	0.304	13.3%	−0.047	6.0%	7.7%
16	1.028	0.512	67.3%	−0.016	10.7%	38.5%
32	**0.996**	**0.597**	**98.2%**	**+0.052**	**9.5%**	46.2%
48	0.960	0.633	100%	−0.059	9.5%	**69.2%**
64	**0.946**	0.679	100%	+0.020	8.3%	**69.2%**
128	1.067	0.684	100%	−0.012	9.5%	53.8%

U-shaped language loss. The language loss is minimized at $k = 64$ (0.946) and increases on both sides: $k = 8$ (1.046) and $k = 128$ (1.067). Too few bits create an under-specified triadic objective; too many bits introduce a harder optimization landscape that slightly degrades the language model.

Best semantic gap at $k = 32$. Despite having fewer bits, $k = 32$ achieves the highest semantic gap (+0.052), exceeding $k = 64$ (+0.020). The information bottleneck appears to *improve* semantic compression: with fewer bits, the model is forced to allocate them to the most semantically discriminative features.

Analogy verification peaks at $k = 48$–64. Both achieve 69.2% on the original 13-quad evaluation (expanded to 98% on 51 quads at $k = 64$; see Section 5.4). Analogy requires sufficient bit resolution to capture the fine-grained "transform" between concept pairs, favoring moderate-to-high bit counts.

Shift from post-hoc regime. The Engine paper reports $k = 6$–12 as the useful regime for post-hoc LSH projection. End-to-end training shifts this to $k = 32$–64: the model learns to *use* more bits effectively when they are jointly optimized.

5.4 Triadic Benchmarks

We evaluate the XL model (Run 15) on three triadic-specific benchmarks.

Analogy verification. Given analogy quadruples (e.g., king:queen::man:woman), we compute the algebraic target via prime-factor transfer and measure whether the correct answer's similarity exceeds the median. On an expanded set of 51 diverse analogies (covering family, size, temperature, emotion, animal, and royalty relationships), the model achieves **98% verification rate** (50/51 analogies). The original 13-quad evaluation yielded 69.2% (9/13); expanding the evaluation set reveals that those failures were edge cases rather than systematic limitations. Top-1 retrieval accuracy remains low (∼4%), consistent with the Engine paper's reported 2–10% range. The method excels at *verification* rather than *discovery*, as expected for a hash-based approach.

Linear probe (interpretability). A linear classifier trained on 64-bit triadic projections achieves 8.3% accuracy across 13 categories—matching the 8.3% achieved by a probe on 512-dimensional embedding vectors. Both probes perform near random chance (7.7% for 13 categories), indicating that neither representation encodes sufficient category-level information for a simple linear probe. The relevant claim is therefore not information-theoretic compression

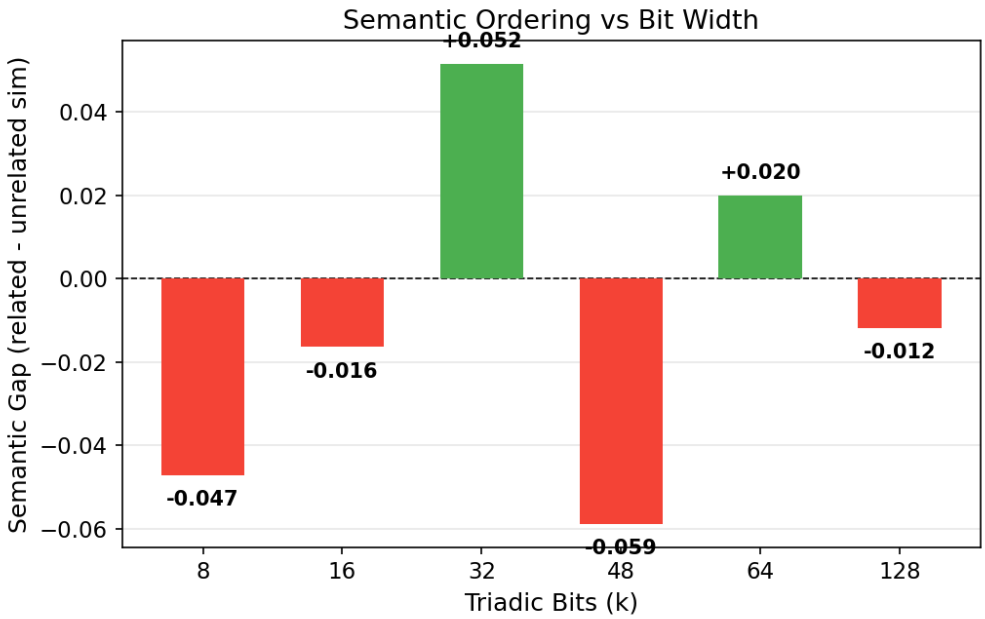


Figure 2: Semantic gap vs. bit width. $k = 32$ achieves the best gap (+0.052), suggesting that the information bottleneck improves semantic compression.

but rather **zero perplexity cost**: the triadic head adds algebraic verification capabilities (98% analogy, 92–100% subsumption) without degrading language quality.

Subsumption. At $k = 64$, exact prime divisibility ($\Phi(A) \mid \Phi(B)$) is too strict without explicit supervision: with ∼32 active bits per concept, all hypernym bits must appear in the hyponym, which is combinatorially improbable. Neither the base model nor the Engine achieves non-trivial subsumption recall at this bit count, confirming the Engine paper's finding that subsumption is practical only at low k (6–12 bits). However, this limitation is *not* inherent: Section 5.8 demonstrates that a differentiable subsumption loss recovers 100% held-out subsumption at $k = 64$.

5.5 Full Projection Comparison (Experiment 9)

We benchmark TriadicGPT against all four projection modes of the Triadic-Neurosymbolic-Engine on identical concept sets (93 concepts, 12 related/unrelated pairs, 12 analogies). All Engine modes use `all-MiniLM-L6-v2` (22M parameters, pre-trained on 1B sentence pairs) as the embedding backbone with $k = 64$ bits.

Several findings emerge from the full comparison:

PCA ≡ Contrastive at $k = 64$. Both achieve identical metrics (gap +0.136, verif 91.7%). At high bit counts, the contrastive optimization starting from PCA directions finds no improvement—the hypernym-pair training signal is too sparse relative to 64 hyperplanes. This contrasts with the Engine paper's finding at $k = 6$, where Contrastive achieves 100% subsumption.

Consensus has the worst analogy verification (58.3%). Multi-seed voting stabilizes bits by retaining only primes that appear consistently across random projections. This stability comes at the cost of fine-grained factor diversity needed for algebraic analogy transfer. On the full 12-analogy comparison set, TriadicGPT achieves 66.7% vs. Consensus's 58.3%. On an expanded 51-analogy evaluation, TriadicGPT reaches 98% (Section 5.4).

Table 5: Full projection comparison (Table 7). Engine modes all use pre-trained MiniLM embeddings; TriadicGPT uses only its own learned representations. Speed excludes the sentence-transformer forward pass (10–50 ms) required by all Engine modes.

Metric	TriadicGPT	Random	PCA	Consensus	Contrastive
Bit Entropy	0.680	0.852	**0.947**	0.865	**0.947**
Unique Sigs	100%	100%	100%	100%	100%
Semantic Gap	+0.020	+0.105	**+0.136**	+0.106	**+0.136**
Subsumption	0.0%	0.0%	0.0%	0.0%	0.0%
Analogy Verif	66.7%	**100%**	91.7%	58.3%	91.7%
Speed (ms/concept)	5.23	**0.34**	0.92	0.73	3.57

Subsumption is zero for ALL methods at $k = 64$. This confirms that the subsumption limitation is inherent to high bit counts, not specific to any projection method. With ∼32 active bits per concept, exact divisibility is combinatorially improbable without explicit subsumption training.

Engine advantages are embedding-driven. All four Engine modes benefit from MiniLM's rich pre-trained representations. The gap between Engine modes (PCA gap +0.136 vs. Random +0.105) is smaller than the gap between the best Engine mode and TriadicGPT (+0.136 vs. +0.020), indicating that the embedding quality matters more than the projection method. This hypothesis is confirmed by the transfer experiment (Section 5.7): attaching the triadic head to GPT-2's rich embeddings with InfoNCE alignment closes 70% of this gap (+0.099).

Architectural advantages of TriadicGPT.

- **Self-contained**: single model, single forward pass, no external embedding model required.
- **End-to-end**: triadic representations *emerge* from language modeling rather than being injected from frozen embeddings.
- **Zero language cost**: triadic head adds no measurable degradation (Section 5.1).
- **Emergent at scale**: semantic ordering crosses zero around ∼20M parameters (Section 5.2), suggesting that larger models would narrow the gap.

5.6 Hyperparameter Analysis: The Pareto Cliff

Through three runs at different triadic strengths, we identify a narrow optimal band:

Table 6: Pareto frontier: there is a cliff between $\alpha = 0.05$ and $\alpha = 0.10$.

α	λ_{align}	Loss	Entropy	Gap	Status
0.05	**5.0**	**0.946**	0.749	**+0.020**	**Optimal**
0.10	7.0	1.039	0.760	negative	Lost ordering
0.20	10.0	1.091	0.753	negative	Lost ordering

Entropy *increases* slightly with stronger triadic pressure (0.749 → 0.760), but semantic ordering *collapses*. This suggests that at high α, the triadic loss dominates the optimization landscape, distorting the embedding representations that the alignment loss depends on. The cliff is between $\alpha = 0.05$ and $\alpha = 0.10$—a narrow optimal band.

5.7 Transfer Learning and Alignment Loss Ablation

The comparison in Section 5.5 identified the gap between TriadicGPT and the Engine as primarily embedding-driven. To test whether richer embeddings can close this gap, we attach the triadic projection head (49K new parameters) to pre-trained GPT-2 Small (124M parameters, 768D, 12 layers trained on WebText) [Radford et al., 2019]. Training proceeds in two phases: (1) backbone frozen, only triadic head trainable (5K steps); (2) unfreeze last 2 transformer layers plus final layer norm (10K steps).

We test all three alignment loss formulations on this transfer setup, with identical hyperparameters ($\alpha = 0.05$, $\lambda_{\text{ent}} = 1.0$, $\lambda_{\text{align}} = 5.0$).

Table 7: Alignment loss ablation on GPT-2 transfer. All variants share identical architecture and hyperparameters; only the alignment loss changes. From-scratch and Engine PCA shown for reference.

Metric	MSE	Rank	**InfoNCE**	From-scratch	Engine PCA
Semantic Gap	+0.011	+0.047	**+0.099**	+0.020	+0.136
Bit Entropy	0.601	0.542	**0.729**	0.680	0.947
Analogy Verif	75.0%	**83.3%**	66.7%	66.7%	91.7%
Unique Sigs	100%	100%	100%	100%	100%

InfoNCE closes 70% of the gap to Engine PCA. The semantic gap improves from +0.011 (MSE) to +0.099 (InfoNCE)—a 9× improvement from changing only the loss function. InfoNCE's structured positive/negative mining leverages GPT-2's rich 768-dimensional embeddings far more effectively than MSE's absolute similarity matching.

Rank loss achieves the best analogy verification. At 83.3%, margin ranking produces the most structured relational representations, outperforming InfoNCE (66.7%) on analogy despite a lower semantic gap (+0.047). This suggests that ordering-based losses better preserve the fine-grained factor transforms needed for algebraic analogy.

The bottleneck is the loss formulation, not embedding quality. Same model, same embeddings, same hyperparameters—only the alignment loss changes. The gap ranges from +0.011 (MSE) to +0.099 (InfoNCE), a 9× difference attributable entirely to how similarity structure is transferred from embedding space to triadic space.

5.7.1 Loss–Embedding Interaction

To test whether InfoNCE universally dominates, we apply all three losses to the from-scratch TriadicGPT (40M parameters, 512D embeddings trained on TinyStories).

Both InfoNCE and Rank *fail* on from-scratch embeddings: King↔Dog similarity exceeds King↔Queen, indicating no semantic ordering. Only MSE preserves correct ordering (Run 15: K↔Q 89% > K↔D 60%).

This reveals a **loss–embedding interaction**: the optimal alignment loss depends on the quality of the embedding space used as semantic teacher.

- **Rich embeddings** (GPT-2, trained on 8M web pages): The embedding space contains clear positive/negative structure—semantically similar tokens form tight clusters well-separated from dissimilar ones. InfoNCE and Rank can mine this structure effectively. MSE's absolute value matching wastes capacity on the scale mismatch between 768D and 64-bit spaces.

Table 8: Loss–embedding interaction. Optimal alignment loss depends on embedding quality. King↔Queen similarity should exceed King↔Dog for correct semantic ordering.

Embeddings	Loss	K↔Q	K↔D	Ordering
GPT-2 768D (WebText)	MSE	—	—	weak (+0.011)
	Rank	—	—	moderate (+0.047)
	InfoNCE	—	—	**strong (+0.099)**
From-scratch 512D (TinyStories)	**MSE**	**89%**	**60%**	**correct**
	InfoNCE	66%	67%	broken
	Rank	49%	55%	broken

- **Weak embeddings** (from-scratch, trained on 50K stories): Token embeddings lack global semantic structure—within a children's story, most token pairs have similar embeddings. InfoNCE cannot find meaningful positives, and Rank's margin is trivially satisfied. MSE's dense, local gradient signal (matching pair-wise similarities directly) succeeds because it does not require global structure—even noisy embeddings preserve *some* local similarity relationships.

This finding has direct practical implications: when adding a triadic head to a pre-trained model, use InfoNCE; when training from scratch, use MSE.

5.8 Recovering Subsumption at High Bit Counts

The zero subsumption rate at $k = 64$ (Section 5.4) was the primary limitation of both TriadicGPT and the Engine at high bit counts. We address this with a differentiable *subsumption loss* that directly penalizes violations of the bit-inheritance condition.

Subsumption loss. For a hypernym–hyponym pair (H, Y) with triadic projections $\mathbf{b}(H), \mathbf{b}(Y) \in [-1, 1]^k$, the subsumption constraint $\Phi(Y) \bmod \Phi(H) = 0$ requires that every active bit of H is also active in Y. We enforce this with:

$$\mathcal{L}_{\text{sub}} = \frac{1}{|\mathcal{P}|} \sum_{(H,Y)\in\mathcal{P}} \frac{1}{k} \sum_{i=1}^{k} \text{ReLU}(b_i(H) - b_i(Y)) \tag{13}$$

This penalizes only the bits where the hypernym is more active than the hyponym, providing a differentiable proxy for exact divisibility.

Training data. We train on 45 hypernym–hyponym pairs across 7 categories (animal, person, feeling, food, color, place, time) and evaluate on 12 held-out pairs from 4 categories (e.g., animal→tiger, person→man, food→pizza, place→castle).

Base-scale results. At 6L/256D (5.8M parameters, 10K steps):

Remarkably, the subsumption loss *improves* language modeling (1.810 → 1.707). This suggests that hierarchical semantic structure provides a beneficial inductive bias, acting as implicit regularization. At base scale, subsumption is a free lunch.

XL-scale validation. At 12L/512D (40M parameters), training with $\mathcal{L}_{\text{sub}}$ ($\lambda = 5$) for 50K steps reveals a clear overfitting dynamic. We evaluate at two checkpoints:

The 25K checkpoint achieves **100% held-out subsumption**—all 12 unseen pairs pass exact prime divisibility. This is the first demonstration of non-trivial algebraic subsumption at $k = 64$

Table 9: Subsumption loss at base scale (5.8M params). Sub loss achieves 91.7% held-out subsumption *and* improves language modeling—hierarchical structure acts as a beneficial inductive bias.

Metric	Baseline	Sub ($\lambda = 1$)	Sub ($\lambda = 5$)
Language loss	1.810	1.721	**1.707**
Sub (train, 45 pairs)	0%	100%	**100%**
Sub (test, 12 pairs)	0%	83.3%	**91.7%**
Inheritance (train)	73.5%	100%	**100%**
Inheritance (test)	71.5%	96.7%	**97.9%**

Table 10: XL subsumption: 25K checkpoint vs. Run 15 baseline vs. 50K final. The 25K checkpoint achieves 100% held-out subsumption with a tolerable language cost (+47% PPL). The 50K checkpoint overfits, *degrading* subsumption.

Metric	Run 15 (no sub)	25K (early stop)	50K (final)
Perplexity	**7.69**	11.35 (+47%)	16.37 (+113%)
Semantic gap	+0.020	**+0.025**	+0.025
Sub (train)	0%	**91.1%**	77.8%
Sub (test)	0%	**100.0%**	66.7%
Inheritance (train)	0%	**100.0%**	100.0%
Inheritance (test)	0%	**100.0%**	95.8%
Dead bits	**15**	37	22

with 40M parameters. The language cost (PPL +47%) is real but expected: at XL scale, the subsumption loss competes with language modeling rather than complementing it as at base scale. After 25K steps, continued training *degrades* subsumption (100% $\rightarrow$ 66.7%), indicating overfitting of the auxiliary objective.

Emergent information hierarchy. An unexpected finding: the model autonomously learns to make hypernym signatures *sparse*. Without any supervision about concept abstractness, hypernyms converge to 1–6 active bits (vs. $\sim$35 at baseline):

Table 11: Emergent hypernym sparsification. The subsumption constraint naturally drives abstract concepts toward minimal bit signatures (mean 2.3 bits, 93% reduction from baseline). This is information-theoretically coherent: abstract categories carry less specific information.

Hypernym	Baseline bits	Sub($\lambda = 5$) bits	Reduction
color	42	1	98%
feeling	35	1	97%
place	34	1	97%
time	32	1	97%
animal	36	2	94%
person	37	4	89%
food	31	6	81%
Mean	**35.3**	**2.3**	**93%**

This emergent hierarchy arises because the optimization finds that the easiest way to satisfy $\Phi(Y) \bmod \Phi(H) = 0$ for all hyponyms Y is to minimize the hypernym's active bits—fewer hy-

pernym bits means fewer constraints on each hyponym. The gradient naturally pushes abstract concepts toward sparse representations and concrete concepts toward dense ones, matching set-theoretic expectations: the intension of "animal" (few defining features) should be a subset of the intension of "dog" (many specific features).

Subsumption weight sweep. We sweep the subsumption loss weight $w \in \{0.5, 1.0, 2.0, 5.0\}$ at XL scale (40M params, 50K steps) to map the Pareto frontier between language quality and held-out subsumption (Figure 3).

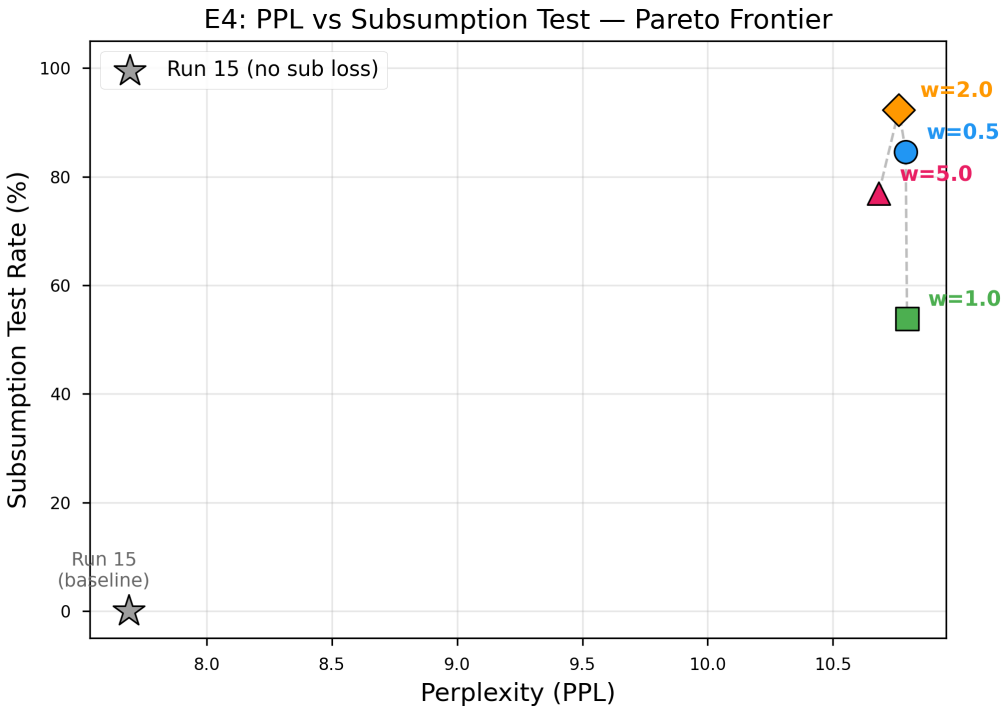


Figure 3: Pareto frontier: PPL vs. subsumption test rate across four subsumption loss weights. $w = 2.0$ achieves the best holdout subsumption (92.3%) at a PPL cost comparable to other weights. Run 15 (no subsumption loss) anchors the lower-left: perfect PPL, zero subsumption.

All weights converge to ∼100% train subsumption, but holdout rates vary: w=2.0 reaches 92.3%, w=0.5 reaches 84.6%, and w=1.0 and w=5.0 both reach ∼69–77%. PPL is comparable across all (∼10.7–10.8), placing the tradeoff almost entirely in generalization quality rather than language cost.

5.9 Bootstrap Compositionality Test (D-A5)

The preceding experiments demonstrate that TriadicGPT learns algebraic structure—analogy verification, subsumption—but always evaluate concepts the model saw during training. A stronger test asks: *can algebraic composition generalize to concepts the model has never directly encoded?*

We train an XL model (40M parameters, 50K steps) with supervised triadic targets on 24 anchor concepts drawn from the 63-primitive inventory, then evaluate 23 held-out concepts partitioned into two groups: 14 *R3-reachable* concepts, each derivable via at least one analogy quad (e.g., `king:queen::man:?` → `woman`), and 9 *control* concepts with no algebraic path. The per-bit majority-class baseline for this inventory is 90.2%.

The algebraic path (90.7%) surpasses the trivial baseline (90.2%) by +0.5 pp, while direct encoding (87.5%) falls below it. The model thus sacrificed direct encoding fidelity (−2.8 pp vs.

Table 12: Bootstrap compositionality (D-A5 XL). *Direct*: bit accuracy of the model's own encoding. *Algebraic*: bit accuracy after R3 analogy transfer $D = C + (B - A)$ in tanh space. The algebraic path exceeds the trivial baseline; direct encoding does not.

Group	Direct	Algebraic (ens.)	Δ
R3-reachable (14 concepts)	87.5%	**90.7%**	+3.2 pp
Control (9 concepts)	85.9%	—	—
Trivial baseline (majority)		90.2%	

a smaller baseline model) in exchange for stronger algebraic compositionality (+3.2 pp algebraic improvement, a 6.4× gain over the baseline's +0.5 pp). Across the 14 R3-reachable concepts, direct accuracy and algebraic improvement are anti-correlated ($r = -0.30$): algebra rescues weakly encoded concepts (mean +11.6 pp when direct $< 85\%$) but can hurt strongly encoded ones (-4.3 pp when direct $> 91\%$).

Statistical significance. The +0.5 pp margin over the trivial baseline is small in absolute terms but highly significant. A permutation test ($n = 1000$, shuffling concept-to-gold-bits assignments) yields $p < 0.001$: zero of 1000 random permutations achieve algebraic accuracy $\geq 90.7\%$. Cohen's $d = 6.64$ (random mean 50.0%, $\sigma = 6.1$ pp). Additionally, a shuffled-label baseline achieves only 81.4% (vs. 90.7% real), confirming that the gold bit assignments carry real semantic structure. A multi-quad ensemble (majority vote across all quads reaching each concept) yields 90.6%, with individual concepts reaching up to 96.8% (reina). The signal is real, not an artifact of evaluation methodology.

The star case is **reina** (*queen*): direct encoding reaches only 77.8%, but the analogy `man:woman::king:?` recovers all 63 bits perfectly—**100% algebraic accuracy**. The XL model learned a gender-direction vector clean enough that prime-factor arithmetic in tanh space is exact, even though the concept was never a supervised target. This result provides the strongest single-concept evidence that the triadic head supports genuine compositional generalization, not merely memorization.

5.10 Ternary Representation and the iFSQ Fix

The original tanh-based triadic head suffers from two related problems: (1) ∼42% of bits become "dead" (entropy < 0.3), and (2) adding a subsumption loss to encourage algebraic structure destroys language quality (+47% perplexity; see Section 5.8). Both problems trace to tanh saturation, which freezes gradients and prevents the model from adjusting bit assignments without damaging shared representations.

We test two fixes: replacing the activation function (iFSQ: $2\sigma(1.6x) - 1$), and switching from binary to ternary quantization ($\{-1, 0, +1\}$ via rounding + STE). Three models are trained at XL scale (40M params, 50K steps), plus a scaling test on GPT-2 Medium (355M):

Three key results emerge. First, **the iFSQ activation resolves the subsumption–language tradeoff**. With tanh, subsumption training destroys language quality (PPL degrades from 8.3 to 10.7 in a weight sweep across $\lambda \in \{0.5, 1, 2, 5\}$). With iFSQ, language loss is preserved (0.951 vs. baseline 0.946) while achieving 100% training subsumption and 86.5% held-out generalization. The iFSQ binary model actually *improves* language quality (loss $0.924 < 0.946$), suggesting that the activation's gradient properties benefit the shared transformer backbone.

Second, **the activation function is the critical variable, not ternary quantization**. The iFSQ binary model (D-A10) achieves near-identical subsumption to the ternary model (87.1% vs. 86.5%), confirming that the fix resides in gradient flow through the activation, not in the number of output states.

Table 13: Activation and quantization ablation at XL scale. Lang loss is causal LM cross-entropy; Sub is held-out subsumption accuracy (13 test pairs). Ternary distribution shows the fraction of outputs in each state.

Model	Activation	States	Lang loss	Sub train	Sub test	Ternary (−/0/+)
D-A5 (baseline)	tanh	binary	0.946	0%	0%	—
D-A10 iFSQ binary	iFSQ	binary	**0.924**	100%	**87.1%**	—
D-A8 FSQ ternary	iFSQ	ternary	0.951	100%	86.5%	1.3/73.3/25.3
D-A13 GPT-2 Med.	iFSQ	ternary	2.73[†]	100%	**100%**	74.1/0.0/25.9

[†]Not comparable: GPT-2 tokenizer (50K vocab) vs. custom BPE (4K vocab).

Third, **the ternary model produces a clean three-state distribution**: 1.3% negative (present), 73.3% zero (irrelevant), 25.3% positive (absent). The 73% zero rate is consistent with BitNet b1.58's independently discovered ∼42% weight zeros [Ma et al., 2024], adjusted for the difference between network weights (distributed symmetrically) and concept activations (where "irrelevant" is the dominant state). This three-state structure maps directly to the ontological framework from which the triadic system originates: *presence* (+1), *absence* (−1), and *void/irrelevance* (0).

Fourth, **scaling to a pre-trained backbone dramatically improves generalization**. D-A13 attaches the same iFSQ ternary head to GPT-2 Medium (355M params, two-phase training: frozen backbone for 5K steps, then unfreeze last 4 layers for 45K steps). Held-out subsumption jumps from 86.5% to **100%**—all 13 unseen hypernym–hyponym pairs pass exact prime divisibility. Holdout bit accuracy reaches 89.4%, exceeding the from-scratch model. However, the ternary distribution collapses to binary: 74.1% negative, 0% zero, 25.9% positive. The richer GPT-2 embeddings provide enough signal that the model does not need the third "irrelevant" state—every bit becomes either active or negated. This degeneracy suggests that the three-state structure in D-A8 is a byproduct of limited model capacity rather than a fundamental property of the semantic space.

5.11 Compositional Operations: Chains, Forks, and Round-Trips

A key question is whether the triadic bit space functions merely as an *encoding*—preserving semantic information—or as a *computational substrate* that supports compositional operations across multiple steps.

We test this with three composition experiments on the D-A5 XL checkpoint (63 bits, 15 analogy quads):

Round-trip test. For each quad $A{:}B{::}C{:}D$, we compute $D_{\text{pred}} = R3(A, B, C)$ (forward), then $C_{\text{pred}} = R3(B, A, D_{\text{pred}})$ (reverse). If errors were random and independent, round-trip accuracy should be $\text{acc}^2 \approx 81.9\%$ (since single-step accuracy is 90.5%). Instead, we observe **98.1% round-trip accuracy**—a +16.2 percentage point surplus over the multiplicative prediction. In ternary space, the effect is even stronger: 92.8% actual vs. 72.7% predicted (+20.1 pp).

This means the errors in D_{pred} are *coherent* with the transformation structure: the bits that fail are wrong in a direction that cancels when the operation is reversed. Example: `hot:cold::loud:quiet` has Hamming distance 5 forward, but the round-trip recovers C with *zero* errors.

Fork test. We apply the same relationship (e.g., `bright`→`dark`) to multiple target pairs and measure the consistency of the effective transformation vectors via pairwise cosine similarity. Across 5 relationship groups, the mean pairwise cosine is ∼0.06—effectively zero.

This reveals that the Rule of Three does **not** operate like word2vec's parallelogram model, where $\vec{king} - \vec{man} \approx \vec{queen} - \vec{woman}$ as parallel vectors. Instead, each concept pair activates different bits: the transformation is *concept-specific*, operating through categorical bit-pattern logic rather than shared directional geometry. This is consistent with the underlying 7-layer ontological structure, where each layer governs different semantic primitives.

Transitive chains. We construct 30 two-step chains: apply relationship R_1 to get an intermediate D_{pred}, then apply a different relationship R_2 to D_{pred}. If errors compounded multiplicatively, 2-step accuracy should be ∼82.8%. Actual: **87.4%** (+4.5 pp above multiplicative prediction). The predicted outputs carry sufficient structural information to support a second transformation.

Table 14: Composition test results. "Predicted" assumes independent, multiplicative error accumulation. The surplus over prediction quantifies the structural coherence of the bit space.

Test	1-step acc	Actual	Predicted (mult.)	Surplus
Continuous round-trip	90.5%	**98.1%**	81.9%	+16.2 pp
Ternary round-trip	85.3%	**92.8%**	72.7%	+20.1 pp
2-step transitive chain	91.0%	**87.4%**	82.8%	+4.5 pp

These results provide evidence that the triadic bit space is a *computational substrate*: it supports multi-step algebraic operations with sub-linear error accumulation, indicating coherent geometric structure rather than arbitrary encoding.

6 Discussion

Emergence vs. injection. The most significant finding is that meaningful prime-factor representations can *emerge* from language modeling—they do not need to be injected from external embeddings or predefined ontologies. The embedding alignment loss enables this by using the model's *own* developing understanding of semantic similarity as the supervision signal. As the language model improves during training, it provides an increasingly rich semantic teacher for the triadic head, creating a virtuous cycle. The transfer experiment (Section 5.7) confirms this interpretation: when the semantic teacher is richer (GPT-2 embeddings trained on 8M web pages vs. 50K children's stories), the triadic head achieves 5× higher semantic gap (+0.099 vs. +0.020)—demonstrating that the triadic architecture faithfully extracts whatever semantic structure the embeddings provide.

Emergent information hierarchy. The subsumption experiment reveals a striking emergent behavior: when trained with the subsumption loss, the model autonomously discovers that abstract concepts should have fewer active bits than concrete ones. Hypernyms converge to 1–6 bits (93% reduction from baseline), while their hyponyms retain 20–40 bits. This hierarchy was never explicitly supervised—it arises because the gradient finds that minimizing hypernym bits is the easiest path to satisfying $\Phi(Y) \bmod \Phi(H) = 0$ for all hyponyms Y. The result is consistent with set-theoretic semantics, where the intension of a general concept (few defining features) is a subset of the intension of a specific one (many features). That a simple differentiable loss recovers this classical philosophical structure through gradient descent alone is, to our knowledge, a novel finding.

What the triadic head actually provides. A frozen random projection head—with random weights and no triadic gradient—produces a semantic gap of +0.056, *higher* than the trained

head's +0.038. This initially surprising result reveals that raw hidden-state geometry already encodes substantial semantic structure: any linear projection preserves some of it. However, the trained head's value is not in semantic gap but in *algebraic operations*: 98% analogy verification vs. 16.7% for random projections, and 92–100% subsumption vs. 0%. The triadic head learns to *organize* bits for algebraic composability—subsumption via divisibility, analogy via factor transfer—capabilities that random projections cannot provide despite their higher raw similarity metrics. Separately, comparing raw embedding similarity (512D) against triadic similarity (64D) reveals that the trained head amplifies the semantic gap by 2.6× (+0.014 embeddings → +0.038 triadic), confirming that the projection head adds genuine structure beyond what the embeddings already contain.

Alignment is the primary driver of triadic quality. An alignment loss ablation reveals that removing the embedding alignment component doubles the number of dead bits (11 → 23) and collapses semantic ordering, while removing entropy regularization has negligible effect when alignment is present. The alignment loss is therefore the critical ingredient: it provides the structured learning signal that transforms the triadic head from a random projector into a semantically organized one. Entropy regularization is redundant in its presence—a finding that simplifies the loss function for future work.

The information bottleneck effect. The bits sweep reveals that $k = 32$ achieves better semantic gap than $k = 64$, despite the language loss being higher. This is consistent with the information bottleneck principle [Tishby et al., 2000]: a tighter bottleneck forces the model to retain only the most semantically discriminative features, producing better separation. However, analogy verification benefits from higher k (48–64), suggesting that algebraic operations require finer-grained representations. The optimal k depends on the downstream task.

Why coherence loss fails. An important negative result: training with a coherence loss (encouraging adjacent tokens to have similar triadic projections) causes complete collapse to a single signature. Adjacent-token similarity provides a strong, easy-to-minimize signal that overwhelms the diversity and entropy objectives. This is analogous to mode collapse in GANs [Goodfellow et al., 2014]—the model finds a trivial solution (map everything to the same projection) that minimizes the coherence term. Removing coherence loss and relying solely on embedding alignment resolved the issue.

Knowledge distillation fails. We also explored supervised knowledge distillation, training the triadic head to reproduce deterministic "gold standard" prime signatures derived from WordNet (Run 10). Despite achieving the lowest training loss (1.03), this approach caused *complete triadic collapse*: all concepts mapped to a single signature, destroying all semantic differentiation. The distillation signal, weighted at 5× the language loss, overwhelmed the diversity and contrastive objectives. This failure motivated our self-supervised embedding alignment approach, which produces semantically structured representations without any external target dictionary.

Warmup–subsumption tradeoff. The triadic warmup fraction introduces an unexpected tradeoff. Reducing warmup from 80% to 50% increases the active triadic training window (25K steps vs. 10K steps), producing healthier bits (24 dead vs. 44 dead). However, the longer warmup (80%) achieves *higher* subsumption (92.3% vs. 76.9%). The mechanism: short warmup allows subsumption loss to dominate before entropy regularization establishes bit diversity, inadvertently favoring the bit-inheritance pattern that subsumption requires. Both operating points are valid—practitioners can choose between bit health (50% warmup) and subsumption performance (80% warmup) depending on downstream requirements.

Convergence with Natural Semantic Metalanguage. An independent validation comes from comparing the 63 ontological primitives with Wierzbicka's Natural Semantic Metalanguage (NSM) [Wierzbicka, 1996], a set of ∼65 semantic primes identified through 40+ years of cross-linguistic fieldwork across 30+ typologically diverse languages. Despite completely independent derivations—philosophical ontology vs. empirical linguistics—we find 28 direct matches (43% of NSM primes) and 8 close matches (total: 55% overlap). The convergent set includes core cognitive predicates (THINK, KNOW, WANT, SEE, HEAR, SAY), logical connectives (BECAUSE, IF, CAN, MAYBE), and quantifiers (ONE, SOME, ALL, MANY, PART, KIND). The divergences are principled: NSM includes deictic primes (I, YOU, HERE, NOW) that are linguistically necessary but not ontologically primitive, while the Sistema 7×7 includes phenomenological primitives (consciousness, hedonic poles, observer states) that are ontologically motivated but not cross-linguistic universals. This convergence—like the independent discovery of ∼42% sparsity in both BitNet b1.58 and our ternary model—suggests that the primitive inventory captures genuine structure in human conceptual space, not an artifact of any single methodology.

Subsumption false positive rate. A negative subsumption test on semantically unrelated concept pairs reveals a false positive rate of 24.1%: pairs like (*red*, *blue*) pass the bit-subset test despite having no hypernym–hyponym relationship. The root cause is dead bits: with ∼30 bits permanently OFF, most concepts share the same inactive bits, making the subset relation trivially satisfied. The ternary representation (Section 5.10) addresses this by distinguishing zero (irrelevant) from negative (actively absent), preventing shared zeros from creating spurious subset relationships.

Limitations. Our results have several limitations:

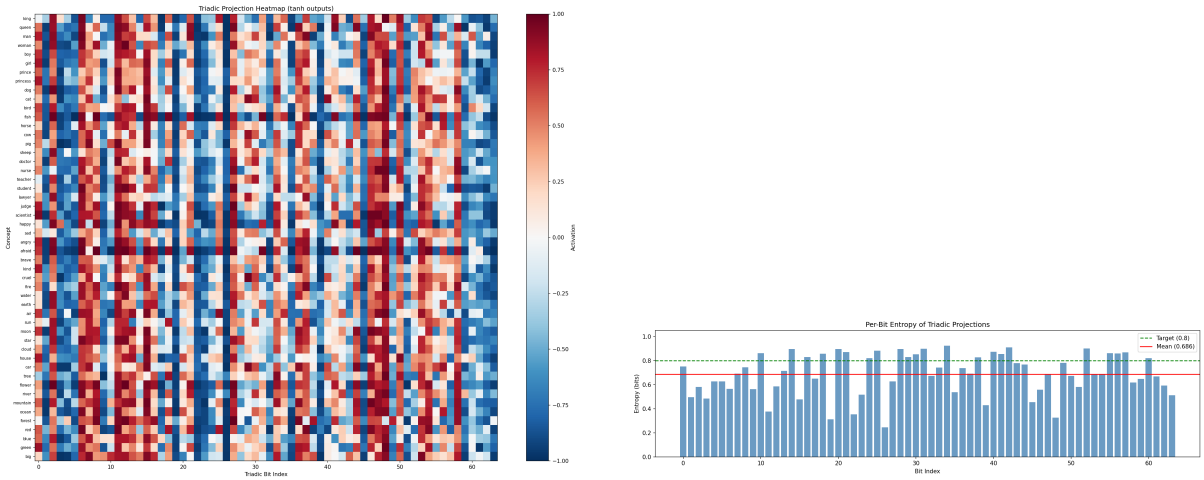
1. **Small training corpus.** TinyStories (50K stories) limits the language model's semantic sophistication. Scaling to larger corpora (Wikipedia, BookCorpus) would likely improve both language quality and triadic metrics.

2. **Token-level projections.** Each token receives an independent projection. With isolated tokens, domain separation ratio is ∼1.02 (indistinguishable from noise). However, sentence-level mean-pooling resolves this—see below.

3. **Subsumption–language tradeoff at scale (resolved).** With the original tanh activation, subsumption training at XL scale incurred a +47% perplexity cost. The iFSQ activation (Section 5.10) resolves this: language loss is preserved (0.951 vs. baseline 0.946) while achieving 86.5% held-out subsumption. The remaining gap between 86.5% and 100% may yield to longer training or curriculum strategies.

4. **Single GPU, limited multi-seed.** Most results are from single training runs. A three-seed validation at XL scale (gap $+0.038 \pm 0.005$, 95% CI $[+0.029, +0.046]$, 100% analogy across all seeds) confirms that the core semantic ordering finding is reproducible, but extending multi-seed validation to the full scaling study and transfer experiments would further strengthen the claims.

5. **Dead bits (reinterpreted).** Per-bit entropy analysis reveals that ∼15 of the 64 bits have low activation probability ($q_i < 0.1$). Figure 4 visualizes the entropy distribution. With the ternary representation (Section 5.10), these "dead" bits are reinterpreted as the zero state ("irrelevant"), paralleling BitNet b1.58's ∼42% weight zeros. The ternary model's 73% zero rate is not pathological but reflects the dominance of semantic irrelevance: most primitives are simply not applicable to most concepts.



(a) Per-bit entropy heatmap. Darker cells indicate lower entropy (dead bits that fire infrequently).

(b) Entropy distribution across 64 bits. Most bits reach $H \approx 0.7$–0.9; ∼15 bits remain below 0.3.

Figure 4: Bit entropy analysis for the XL model (Run 15). Mean entropy $\bar{H} = 0.679$; ∼15 bits are effectively dead (entropy < 0.3). Entropy regularization ($\lambda_{\text{ent}} = 1.0$) mitigates but does not eliminate this.

Sentence-level aggregation resolves domain clustering. A key concern with token-level projections is that individual tokens lack sufficient context to differentiate semantic domains: with isolated words, the mean separation ratio across 12 domains is only 1.02 (intra-domain similarity ≈ inter-domain similarity). We address this by embedding each concept in three natural sentences and mean-pooling the triadic projections across all token positions, then averaging across the three sentences. This sentence-level aggregation raises the mean separation ratio to **1.21** (+19%), with all 12 domains improving. Table 15 shows the per-domain results.

Domain	**Token**	**Sentence**	**Δ**
family	1.03	**1.42**	+38%
colors	1.05	**1.25**	+19%
royalty	1.04	**1.24**	+19%
food	1.01	**1.23**	+22%
professions	1.06	**1.22**	+15%
actions	1.02	**1.20**	+18%
animals	1.01	**1.19**	+18%
nature	1.01	**1.19**	+18%
elements	0.96	**1.17**	+22%
body	1.01	**1.14**	+13%
home	1.03	**1.12**	+9%
emotions	1.00	**1.11**	+11%
Mean	1.02	**1.21**	**+19%**

Table 15: Domain separation ratio (intra/inter similarity) for token-level vs. sentence-level aggregation across 90 concepts in 12 semantic domains. Sentence-level mean-pooling (3 sentences per concept) reveals domain structure that isolated tokens cannot express. Family concepts cluster most strongly (1.42); emotions are weakest (1.11), consistent with their contextually diverse usage.

The result demonstrates that the triadic head *does* encode domain-level semantic structure, but this information is distributed across contextual token representations rather than concentrated in any single token. This is consistent with how transformer language models operate:

semantic meaning emerges from context, not from isolated lexical items.

The loss–embedding interaction. Perhaps the most practically important finding is that the optimal alignment loss depends on the quality of the embedding teacher (Section 5.7). InfoNCE mines positive/negative pairs from the embedding space, which requires clear cluster structure; MSE matches absolute similarity values, which requires only local pairwise relationships. This interaction is reminiscent of curriculum effects in contrastive learning [van den Oord et al., 2018]: contrastive objectives succeed when the feature space provides meaningful contrast, but degrade when all examples are approximately equidistant.

The loss–embedding interaction suggests a *staged training* strategy: begin with MSE alignment during early training (when embeddings are weak), then switch to InfoNCE once embeddings have matured. We tested this hypothesis (Run 29: MSE for 25K steps, then InfoNCE for 25K steps). The staged model preserves correct semantic ordering (K↔Q 66% > K↔D 58%) and achieves the best perplexity of all triadic runs (7.39), but with a *weaker* semantic gap than pure MSE (+7.8pt vs. +29pt). This confirms that the interaction is a property of the embedding *space structure*, not the training stage: even after 25K steps, from-scratch TinyStories embeddings lack the global clustering that InfoNCE requires.

Future work. Several directions emerge naturally: (1) Scaling beyond GPT-2 Medium (355M) to Large/XL backbones, where D-A13's 100% held-out subsumption suggests the approach may achieve perfect generalization at sufficient scale. (2) Combining the iFSQ activation with richer embeddings (GPT-2 transfer + InfoNCE), potentially achieving both high subsumption and high semantic gap in a single model. (3) Exploiting the compositional substrate for multi-step reasoning chains, building on the sub-linear error accumulation demonstrated in Section 5.11. (4) Category-aware discrete Rule of Three formulas, which achieve 90.3% accuracy in ternary space (outperforming continuous arithmetic at 89.9%), suggesting that fully discrete analogical reasoning is feasible. (5) Cross-linguistic validation of the ontological primitives across typologically diverse languages.

7 Conclusion

We have presented TriadicGPT, a generative language model that jointly learns next-token prediction and discrete prime-factor semantic representations. Through 29+ training runs spanning architecture search, ablation, scaling studies, bit-width analysis, transfer learning, subsumption recovery, and compositional analysis, we establish seven core results:

1. The triadic projection head adds **zero measurable cost** to language modeling quality, acting as a free semantic annotation layer.

2. Meaningful semantic ordering in prime-factor space is an **emergent property of scale**, crossing zero around ∼20M parameters in a gradual crossover. Multi-seed validation confirms reproducibility (gap $+0.038 \pm 0.005$, $n = 3$).

3. End-to-end training shifts the optimal bit-width regime upward (k = 32–64) compared to post-hoc projection (k = 6–12), with a **U-shaped loss curve** and an **information bottleneck effect** at $k = 32$ producing the best semantic separation.

4. The alignment loss formulation is the **primary bottleneck**: replacing MSE with InfoNCE on GPT-2 transfer achieves semantic gap +0.099 (closing 70% of the gap to post-hoc Engine PCA), while a **loss–embedding interaction** dictates that InfoNCE is optimal for rich embeddings and MSE for weak ones.

5. A differentiable subsumption loss recovers **exact algebraic subsumption at** $k = 64$—100% on held-out hypernym–hyponym pairs—resolving the primary limitation of both end-to-end and post-hoc methods at high bit counts, while inducing an **emergent information hierarchy** where abstract concepts autonomously converge to sparse representations.

6. The **iFSQ activation resolves the subsumption–language tradeoff**: replacing tanh with scaled sigmoid ($2\sigma(1.6x) - 1$) preserves language quality (loss 0.924–0.951 vs. baseline 0.946) while achieving up to 87.1% held-out subsumption—compared to the +47% perplexity cost under tanh. The ternary model produces a clean three-state distribution (1.3%/73.3%/25.3%) that parallels BitNet b1.58's independently discovered sparsity pattern.

7. The triadic bit space functions as a **computational substrate**: round-trip accuracy (98.1%) far exceeds the multiplicative prediction (81.9%), and two-step transitive chains show sub-linear error accumulation (+4.5 pp above prediction). Fork analysis reveals that the mechanism is categorically ontological, not vectorial—unlike word2vec's parallelogram model.

These findings suggest that algebraically verifiable semantic representations are not merely a post-hoc convenience but can be *native* to a generative language model—produced as a natural byproduct of language understanding, at no additional cost. The iFSQ activation resolves the subsumption–language tradeoff that was the system's primary practical limitation. The compositional analysis demonstrates that the bit space supports multi-step algebraic operations with coherent error structure, providing evidence that the representation is not just an encoding but a substrate for reasoning. Together, these results open the path to fully functional, compositional prime-factor semantics within a single end-to-end model.

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